

TrES-5b

R-1478

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_130528 · created 2026-08-01T12:34:40+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456440.803 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.04
Transit depth (Rp/Rs)^2	2.97 +/- 1.4 [%]
Orbital Inclination (inc)	83.21 +/- 2.2
Airmass coefficient 1 (a1)	1.733 +/- 0.051
Airmass coefficient 2 (a2)	-0.282 +/- 0.023
Scatter in the residuals of the lightcurve fit is	2.92 %
Best Comparison Star	None
Optimal Aperture	2.51
Optimal Annulus	11.29
Transit Duration (day)	0.064 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.04 vs 0.142 (deviation 0.75σ)
Duration fit vs published	0.064 d vs 0.0758 d (deviation 0.62σ)
Mid-time O–C	-3.3 min
Depth SNR	4.3
Residual scatter	2.92 %

Light curve

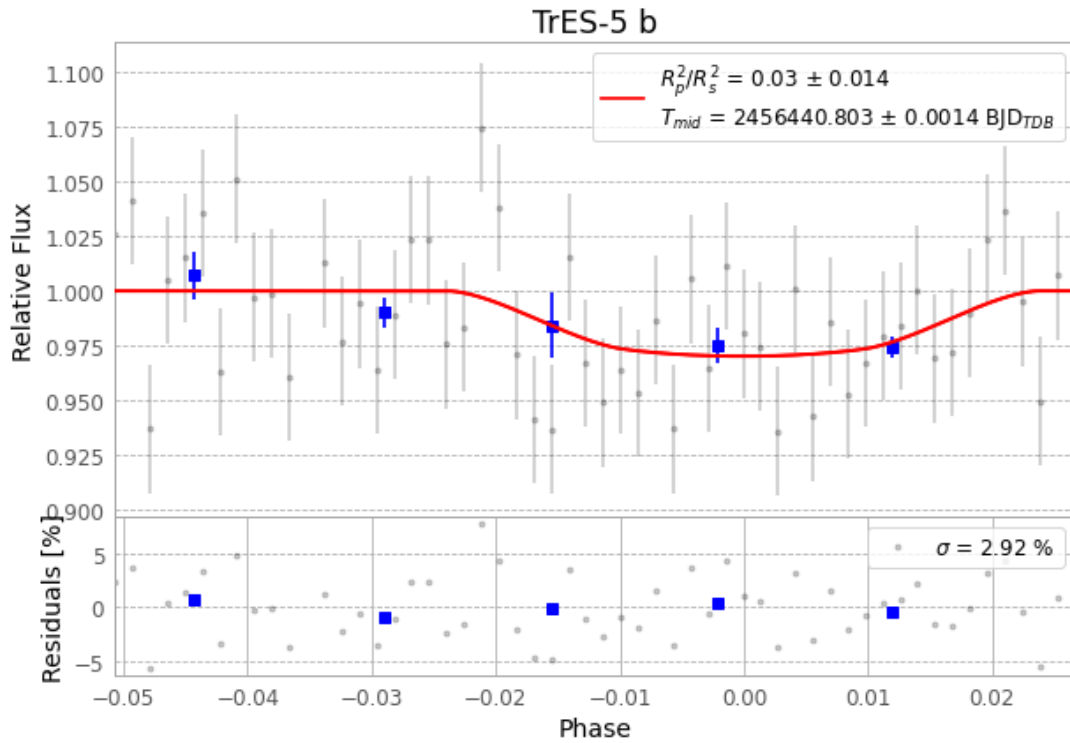


Figure 1 — FinalLightCurve_TrES-5 b_2013-05-27.png (SHA-256 f510d969ac57ed91...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 131], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7f482fe053083255...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

57 FITS frames (37 MB), TRES-5130528052412.FITS → TRES-5130528081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-130528.log (71f960f18fbc...), FinalLightCurve_TrES-5 b_2013-05-27.png (f510d969ac57...), FinalLightCurve_TrES-5 b_2013-05-27.csv (fe41694ace2d...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 12:34Z.